Analysis of Non-Invasive Physiological Markers in Heart Disease Diagnosis

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Disclaimer: The analysis, codebase, and visualizations presented in this document were developed independently for academic purposes using a public dataset from Kaggle. The reported performance metrics are intended solely for exploratory evaluation and should not be interpreted as validated clinical diagnostic guidance.

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1 Introduction

Cardiovascular diseases (CVDs) are the leading cause of global mortality. Early detection via non-invasive physiological metrics provides a critical window for clinical intervention. This project aims to build an end-to-end analytical pipeline to process clinical metrics, handle missing data, train classification algorithms and detect physiological anomalies to predict heart disease diagnosis.

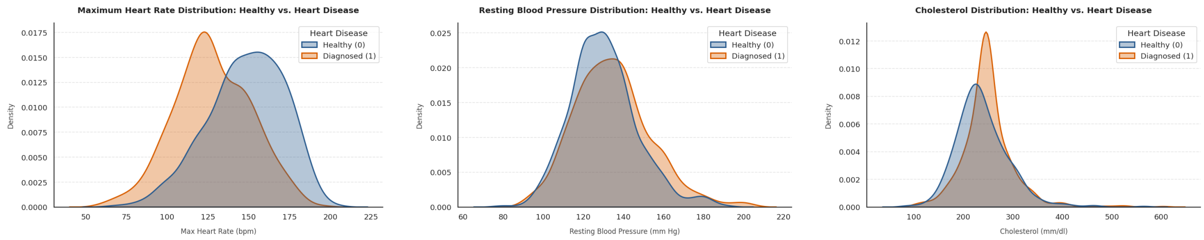
2 Data Extraction & Cleaning

Raw clinical records were loaded into a `SQLite3` for structured querying. While no missing records were found in any column, 1 entry for Resting Blood Pressure (BP) and 172 entries for Serum Cholesterol ($Chol$) contained zero-values representing missing data. These missing values were subsequently imputed using median values grouped by age categories to preserve baseline population distributions.

$$x_{clean} = \begin{cases} \text{Median (Grouped By HeartDisease)}, & \text{if } x = 0 \text{ for } x \in \{BP, Chol\} \\ x, & \text{otherwise} \end{cases} \tag{1}$$

3 Exploratory Data Analysis

Continuous variables were analyzed to evaluate distribution differences between healthy and diagnosed patient cohorts.



(a) Max. HR by Diagnosis (b) Resting BP by Diagnosis (c) Cholesterol by Diagnosis

Figure 1: Comparative density distributions of key physiological metrics.

3.1 Statistical Hypothesis Testing

Subsequently, hypothesis testing was used to test whether maximum heart rate, resting blood pressure and cholesterol of heart disease patients was less than that of healthy patients.

A two-sample t-test was used such that the null hypothesis was 'There is no significant difference between the values of the respective physiological feature between heart disease patients and healthy controls':

$$t = \frac{\bar{X}_{\text{disease}} - \bar{X}_{\text{healthy}}}{\sqrt{\frac{s^2_{\text{healthy}}}{n_{\text{healthy}}} + \frac{s^2_{\text{disease}}}{n_{\text{disease}}}}} \tag{2}$$

Where:

- $\bar{X}_{\text{disease}}$ and $\bar{X}_{\text{healthy}}$ are the sample means of maximum heart rate (`maxhr.mean()`).
- s^2_{disease} and s^2_{healthy} are the sample variances (`std()**2`).
- n_{disease} and n_{healthy} are the respective sample sizes (`len()`).

Using a left-tailed p-test, it was found that there was no significant difference between the two groups for resting blood pressure (p=0.99986) and cholesterol (p=99915) where a threshold p-value of 0.05 was used, however the difference in maximum heart rate values was statistically significant (p = 5.32086e-37), meaning heart disease patients had lower maximum heart rates compared to healthy controls.

4 Classification

I evaluated 2 classification models, namely Random Forest and Logistic Regression, prioritizing **Recall** (Sensitivity) over simple accuracy to minimize false negatives in a medical screening context:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

The objective of each classification model was to correctly predict whether the patient had heart disease or not from the 11 predictive parameters in the dataset. Parameters like test_size were standardized across algorithms to ensure fairness.

4.1 Results

Table 1 summarizes the performance across evaluated baseline models.

Table 1: Model Performance Metrics

Model	Accuracy	Recall	F1-score
Logistic Regression	0.88	0.89	0.88
Random Forest	0.89	0.91	0.90

4.2 Confusion Matrix

The confusion matrix for the Random Forest model is as shown in Figure 1 below.

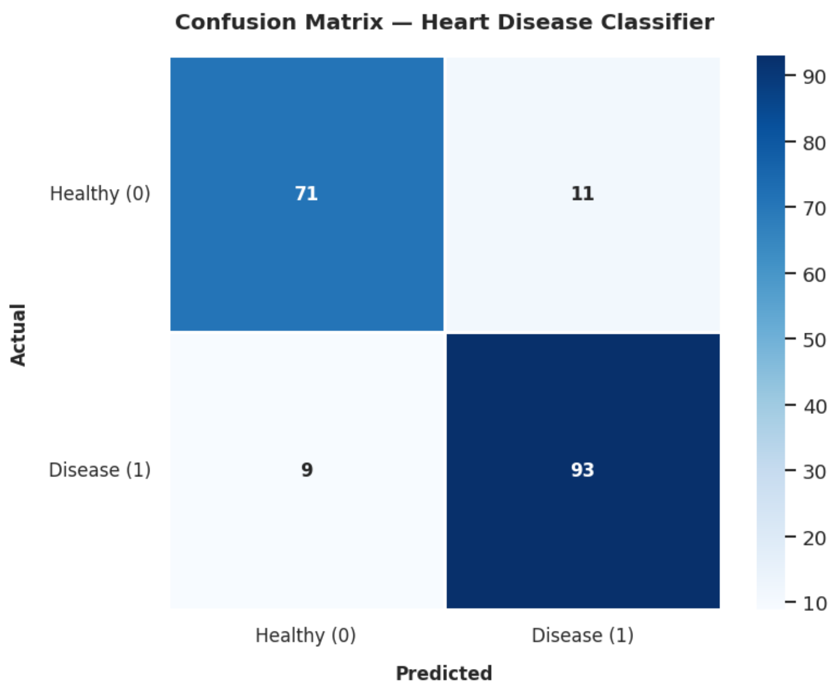


Figure 2: Confusion Matrix visualizing the Random Forest model's result.

4.3 Feature Importance

Utilizing the built-in feature_importances_ function in RandomForestClassifier, the respective Gini importance of each feature used to in the classification model were ranked:

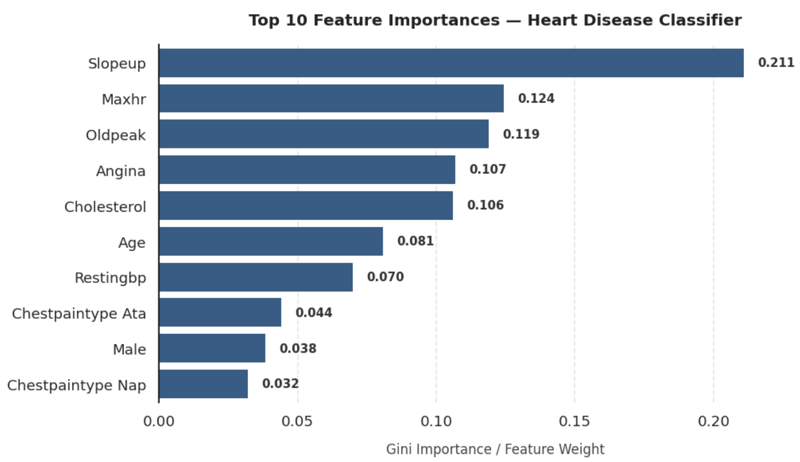


Figure 3: Ranked Importance of each Feature by Gini Importance.

5 Anomaly Detection

The Local Outlier Factor algorithm was used in anomaly detection, detecting anomalous entries based on the continuous physiological markers (age, resting blood pressure, cholesterol, fasting blood sugar, maximum heart rate and oldpeak).

Since anomaly detection is a type of unsupervised learning, the tradeoff between anomaly percentile cutoff and diagnostic concordance was first analyzed:

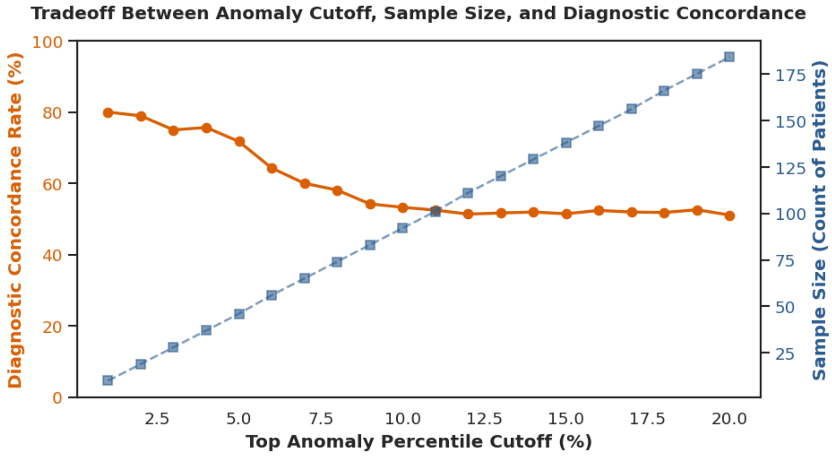


Figure 4: Tradeoff between Percentile Cutoff and Diagnostic Concordance.

From the figure above, the anomaly percentile cutoff was taken to be 2%, flagging 19 anomalies from the dataset. The figure below shows the distribution of Local Outlier Scores, categorized by the heart disease diagnosis:

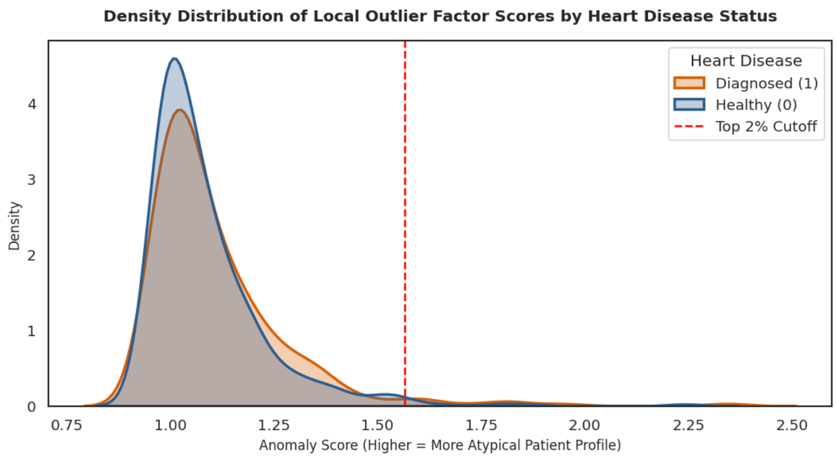


Figure 5: Tradeoff between Percentile Cutoff and Diagnostic Concordance.

The KDE plot shows that the anomaly score distribution for patients diagnosed with heart disease (HeartDisease = 1) is right-shifted compared to healthy control patients. By isolating the top 2% most anomalous records, we identified a specific high-risk subgroup characterized by:

- **Extreme Cholesterol Levels:** Average serum cholesterol of 350.3 mg/dL vs. 242.3 mg/dL in the baseline population.
- **Diagnostic Concordance:** 73.7% of the top-ranked anomalous profiles belonged to patients diagnosed with heart disease, validating that physical extremity strongly correlates

with actual cardiac pathology.

6 Conclusion & Improvements

From the given sample dataset, this analysis confirms that physiological extremity (in continuous markers from the dataset) strongly correlates with heart disease condition. It has also demonstrated the strength of using a Random Forest model for this specific dataset in achieving greater overall sensitivity of 91%.

Due to the limited sample size of the dataset, which mainly comprises of samples from the United States, Hungary and Switzerland, the findings may not be applicable to other demographic groups. Finally, the findings only show that there is a strong correlation between given factors and heart disease, however, there may be confounding variables which caused or strengthened the correlation.